Fishery conditioning and seasonal catch allocation for the shortcut MSE of ane.27.9aS

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Stock: Anchovy (*Engraulis encrasicolus*) in the southern part of Division 9.a (Gulf of Cadiz and southern part of Portugal), ane.27.9aS

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1 Fishery conditioning

1.1 General approach

The fishery component of the Operating Model (OM) was constructed from the conditioned ane.27.9aS trajectories described in WD01. The fishery is represented in FLBEIA by one fleet (`SEINE`), one métier (`ALL`), and one exploited stock (`ANE`) (Garcia *et al.*, 2017).

Historical fishery variables were transferred from the SS3 bootstrap realizations, maintaining the correspondence between the population and fishery of each OM. The historical period covers 1989–2024 and the projection covers 2025–2054. The 100 historical conditions were expanded to 1,000 trajectories, with 10 replicates per OM.

For the projection, catchability, fleet variables, and seasonal catch allocation were parameterized from recent historical periods. Effort, fishing mortality, and realized catches during the projection are subsequently generated within the FLBEIA simulations.

The reference seasonal allocation was defined from the quarterly catch proportions observed during 2015–2024. Historical variation in these proportions was examined to define alternative patterns subsequently used in the robustness analysis.

2 Methods

2.1 Historical fishery structure

The historical fishery remains associated with each biological OM condition. Thus, differences among OMs are not limited to population state and dynamics; each OM also retains the catch history corresponding to its SS3 bootstrap refit.

The 100 historical conditions were expanded to 1,000 trajectories, maintaining the correspondence between each projection trajectory and its source OM. Numbers and weights at age for landings and discards were transferred to the FLBEIA fishery component through an `FLCatchExt`, integrated into métier `ALL` and fleet `SEINE`.

In SS3, the weight at age used for the stock and catch is the same. This assumption was retained during conversion to FLBEIA by checking, for each OM, the correspondence between `landings.wt` and the weight at age of the biological component (`wt`). A maximum absolute-difference tolerance of 10^{-10} was used.

2.2 Fleet variables and projection assumptions

The projected fishery retains the historically conditioned fleet structure, whereas effort and catches are determined dynamically during the simulation. The fleet object includes the `effort.share`, `vcost`, `effort`, `capacity`, `crewshare`, and `fcost` components.

For 2025–2054, `landings.wt`, `discards.wt`, `vcost`, `fcost`, and `crewshare` were fixed from the iteration-specific 2022–2024 mean. Thus, the conditions used during the projection retain differences among OMs derived from historical conditioning.

In the configuration used, `vcost` and `effort` were set to 1, `capacity` to 5,000, and `crewshare` and `fcost` to 0. `effort.share` was retained from the input file. `alpha`, `beta`, `effort.share`, and `capacity` were initially extended from historical values. During the projection, `beta` and `effort` were initialized at 1, whereas `landings.n` and `discards.n` were initialized at 0.

These values define the initial fishery conditions. Effort, fishing mortality, and realized catches during 2025–2054 are subsequently determined within the FLBEIA simulations.

2.3 Effort and catch models

During the projection, the fishery responds to the target catch established by the management procedure through adjustment of fishing effort. The relationship between effort and stock exploitation is represented through catchability (`catch.q`), whereas realized catch is calculated from stock abundance and natural and fishing mortality using the Baranov catch equation.

In FLBEIA, this dynamic is implemented using `SMFB` as the effort model and `Baranov` as the catch model (Garcia *et al.*, 2017). The effort constraint is linked to stock `ANE` and specified in terms of catch.

Catchability (`catch.q`) was calculated for each OM using the 2020–2024 period and kept constant during 2025–2054. Therefore, differences among OMs in this component are retained during the projection. Non-finite values obtained when calculating `catch.q` were replaced by zero.

Fleet capacity was kept constant through `fixedCapital`, with `capacity = 5000`. `catch.threshold` was set to 0.9 and `alpha` to 1.

Catch was assigned to landings (`landings.sel = 1`) and discards were not modeled (`discards.sel = 0`). The discard mechanism associated with catches above quota was also disabled (`discard.TAC.OS = FALSE`).

2.4 Reference seasonal catch allocation

The OM is structured into four quarters and, therefore, annual catch must be distributed among the four seasons. This allocation determines when exploitation occurs within the year and its temporal relationship with population dynamics, including spawning in Q2 and recruitment entry in Q3.

Seasonal allocation is expressed as the proportion of annual catch occurring in each quarter. For OM i, year y, and quarter s, this proportion is calculated as

$$p_{iys} = \frac{C_{iys}}{\sum_{s=1}^{4} C_{iys}},$$

where C_{iys} is the biomass caught in quarter s. The four proportions sum to one for each OM-year combination.

For the reference configuration, the seasonal allocation of each OM was defined as the mean of the quarterly proportions observed during 2015–2024:

$$\bar{p}_{is}^{2015:2024} = \frac{1}{10} \sum_{y=2015}^{2024} p_{iys}.$$

The resulting four proportions were retained during 2025–2054 for the corresponding OM. Thus, this parameterization determines the distribution of catch among quarters, but not the magnitude of annual catch, which is determined during the simulation.

Each of the 100 OMs retains its own 2015–2024 seasonal allocation. The ten replicates derived from the same OM share these proportions. For historical diagnostics, one representative trajectory per OM was used, avoiding treatment of replicates as independent historical observations.

Implementation was checked by reconstructing the mean 2015–2024 proportions from historical catches and comparing them with the values stored in `fleets.ctrl`. A maximum absolute-difference tolerance of 10^{-10} was used.

2.5 Historical catch by age and quarter

The age composition of catch describes which ages contribute to exploitation in each quarter and allows its variation during the historical period to be characterized.

For each OM, year, and quarter, catch at age was calculated as the sum of landed and discarded numbers. The numbers caught at each age were then expressed as a proportion of the total catch in the quarter.

These proportions were used only to describe the historical age composition of the conditioned fishery. They were not used to fix catch composition during 2025–2054. During the projection, catch at age is determined dynamically from population structure and simulated exploitation in FLBEIA.

2.6 Diagnostics of temporal variation in seasonal allocation

The distribution of catches among quarters may vary over time, changing when exploitation occurs within the year. Because the reference configuration uses the mean 2015–2024 allocation, the historical series was examined to characterize its temporal variation and determine whether patterns different from that used in REF exist.

Diagnostics were based on the annual mean quarterly proportions of the 100 historical OMs. For each quarter, change in the catch proportion over time was evaluated using a linear regression against year. Changes in the series were also examined using `strucchange::breakpoints()` and `changepoint::cpt.meanvar()`, the latter using PELT with an MBIC penalty.

The distribution of catch among the four quarters was also analyzed jointly. Principal component analysis (PCA) was applied to centered and scaled annual quarterly proportions to summarize the main changes in seasonal allocation. Hierarchical clustering was then applied to the scaled proportions, using Euclidean distance and Ward's method (`ward.D2`), to group years with similar seasonal distributions.

Three groups ($k = 3$) were used to summarize contrasting historical patterns and define alternative seasonal-allocation configurations. These groups are used as a tool to characterize historical variability and are not interpreted as three discrete fishery regimes.

These diagnostics do not modify the reference configuration. The identified historical patterns are subsequently used to define alternative seasonal-allocation configurations in the sensitivity/robustness analysis.

2.7 Seasonal patterns for the robustness analysis

Alternative patterns for distributing annual catch among the four quarters were defined from historical variability. These patterns represent different timings of exploitation within the year without, by themselves, modifying the magnitude of annual catch.

The patterns were obtained from the mean of the full historical period, the 2015–2024 mean, the 2022–2024 mean, and the means of the three groups identified by clustering. Each vector was normalized so that the four quarterly proportions summed to one.

Table 1: Seasonal catch allocation used in the reference projection. Values assigned to each OM correspond to its own 2015–2024 mean; the table summarizes these OM-specific values for the 100 historical OMs.

Quarter	Mean reference share	SD among OMs
1	0.139	0.002
2	0.416	0.004
3	0.335	0.004
4	0.111	0.002

Table 2: Historical seasonal catch-allocation proportions for the 100 conditioned OMs and years 1989–2024.

Quarter	Mean	SD	5%	Median	95%
1	0.200	0.095	0.024	0.200	0.385
2	0.415	0.116	0.231	0.405	0.635
3	0.283	0.099	0.087	0.294	0.427
4	0.102	0.075	0.005	0.089	0.268

The reference configuration uses the 2015–2024 mean. For the sensitivity analysis, the patterns corresponding to clusters 1 and 2, denoted S-C1 and S-C2, were also selected. These patterns retain the historical structure observed in each group and allow the effect of modifying the within-year temporal distribution of exploitation to be evaluated subsequently.

WD03 documents the construction and historical origin of these patterns. Their implementation and effects on MSE performance are evaluated in WD08.

3 Results

3.1 Reference seasonal allocation

The 2015–2024 seasonal means reconstructed from historical catches matched the values stored in the reference fleet control object. The maximum absolute difference was 0e+00, compared with the implementation criterion of 10^{-10}.

Across the 100 historical OMs, the mean reference allocation was Q1 = 0.139; Q2 = 0.416; Q3 = 0.335; Q4 = 0.111 (Table 1).

3.2 Historical variation in seasonal allocation

The distribution of exploitation among quarters varied during 1989–2024. Table 2 summarizes the catch proportions observed across OMs and years and describes the magnitude of this variation in each quarter.

Temporal diagnostics describe how the contribution of each quarter to annual catch changed during the historical period. Table 3 presents the linear trends and changes identified in the quarterly series.

These results characterize historical variation in the temporal distribution of exploitation. The reference configuration retains the OM-specific 2015–2024 mean.

3.3 Multivariate seasonal patterns

The first two principal components explained, respectively, 54.9% and 32.5% of the variation in the scaled quarterly proportions, and 87.3% jointly.

Hierarchical clustering with $k = 3$ grouped years with similar quarterly distributions and produced three mean seasonal-allocation patterns. These groups summarize different historical configurations of the temporal

Table 3: Temporal diagnostics of historical seasonal catch allocation. The diagnostics characterize the historical series and are not used to redefine the reference OM.

Quarter	Slope	p-value	R2	Breaks	Break years	Changes	Change years
1	-0.005	0.001	0.288	1	1995	0	NA
2	0.000	0.813	0.002	0	NA	0	NA
3	0.005	0.001	0.281	1	1994	1	1994
4	-0.001	0.653	0.006	0	NA	0	NA

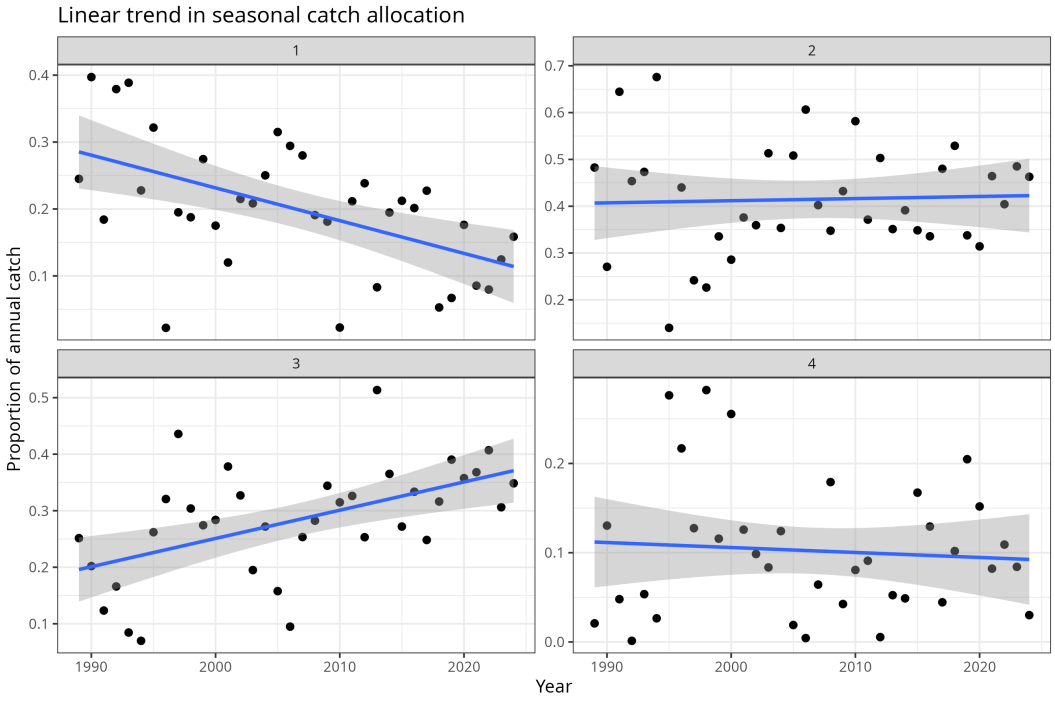


Figure 1: Annual seasonal catch-allocation proportions and quarter-specific linear trends during 1989–2024. The diagnostic is based on annual mean proportions across the 100 conditioned OMs.

Table 4: Mean quarterly catch-allocation proportions for the three diagnostic historical clusters.

cluster	Q1	Q2	Q3	Q4
1	0.271	0.534	0.164	0.031
2	0.235	0.309	0.299	0.157
3	0.099	0.438	0.364	0.098

distribution of exploitation and were used to construct patterns for the robustness analysis. They are not interpreted as three chronological fishery regimes.

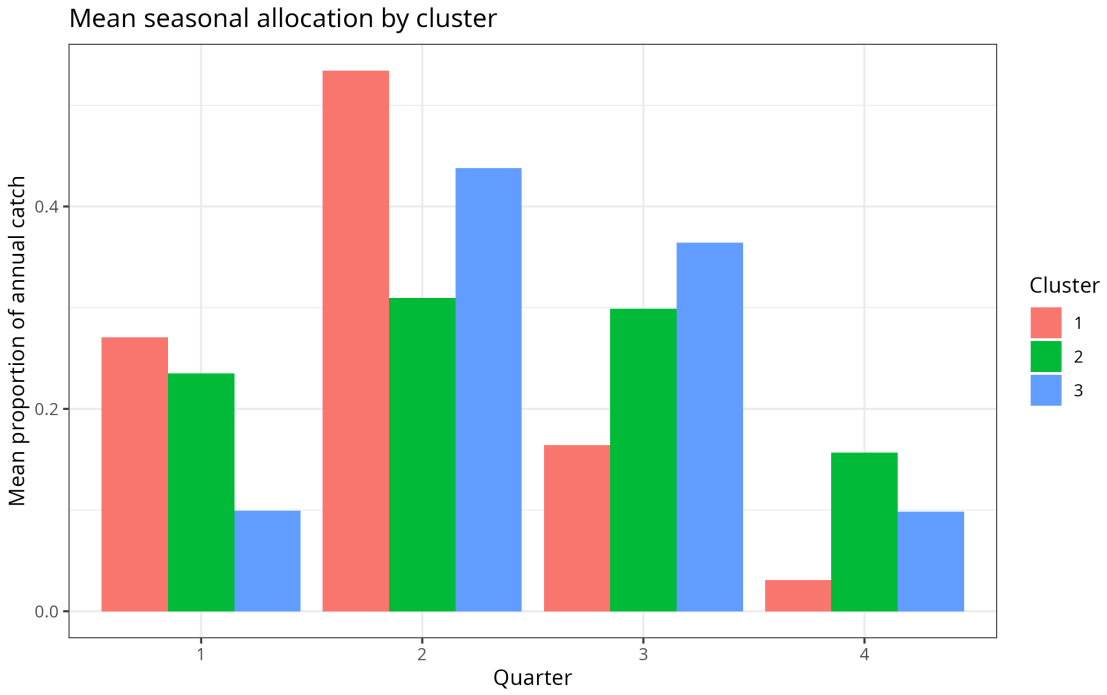


Figure 2: Mean quarterly catch-allocation proportions for the three historical groups. The groups are used to construct alternative seasonal-allocation patterns and do not modify the reference OM.

3.4 Seasonal patterns for the robustness analysis

Six seasonal-allocation patterns were obtained from the historical diagnostics: the 1989–2024 mean, the 2015–2024 mean, the 2022–2024 mean, and the means of the three groups identified by clustering.

Each pattern represents a different distribution of annual catch among Q1–Q4. The reference configuration retains OM-specific 2015–2024 means. Therefore, the aggregated `recent10` pattern is used only as a summary of this period and does not replace the OM-specific proportions used in REF.

The patterns corresponding to clusters 1 and 2 are subsequently used as S-C1 and S-C2 in the robustness analysis documented in WD08.

3.5 Historical catch by age and quarter

Catch age composition differed among quarters during the historical period. Table 6 summarizes the mean contribution of each age to quarterly catch in the conditioned OMs.

These proportions describe the age structure of historical exploitation. They are not used as a fixed

Table 5: Candidate seasonal catch-allocation patterns generated from the historical diagnostics. The reference OM retains OM-specific 2015–2024 means rather than replacing them with the single aggregated recent10 vector shown here.

Scenario	Q1	Q2	Q3	Q4
historical	0.200	0.415	0.283	0.102
recent10	0.139	0.416	0.335	0.111
recent3	0.121	0.451	0.354	0.074
cluster_1	0.271	0.534	0.164	0.031
cluster_2	0.235	0.309	0.299	0.157
cluster_3	0.099	0.438	0.364	0.098

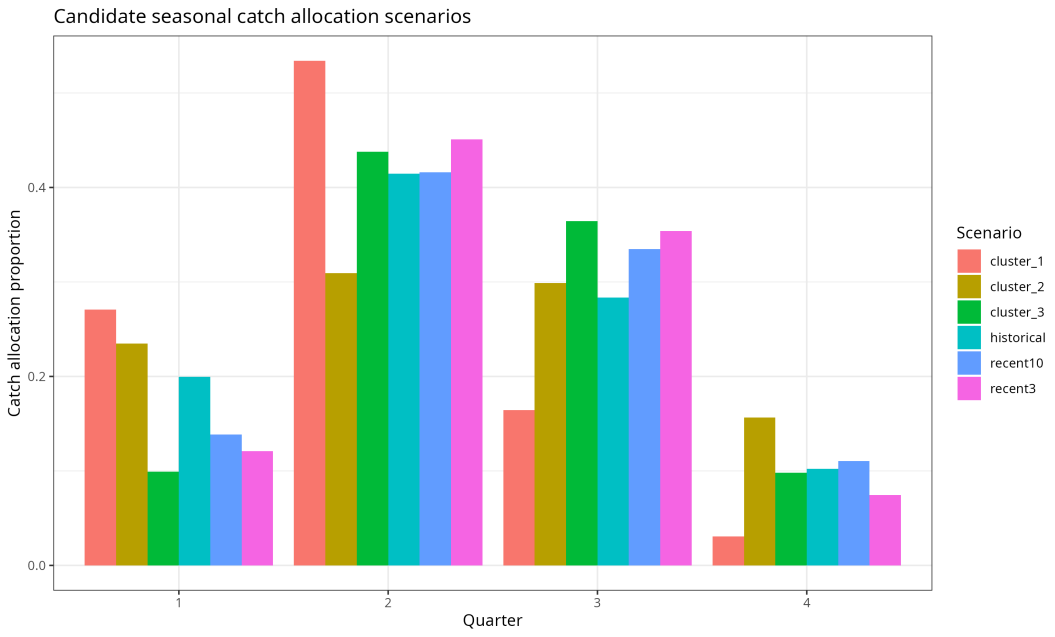


Figure 3: Candidate seasonal catch-allocation patterns generated from the historical diagnostics. These patterns define possible sensitivity scenarios; their effects on MSE performance are evaluated subsequently.

Table 6: Mean historical catch composition by age and quarter across conditioned OMs and historical years. Age 3 represents the 3+ group.

season	Age_0	Age_1	Age_2	Age_3plus
1	0.000	0.872	0.123	0.005
2	0.000	0.916	0.080	0.004
3	0.266	0.673	0.059	0.003
4	0.583	0.374	0.041	0.002

composition during the projection, in which catch at age is determined from the available population and simulated exploitation in FLBEIA.

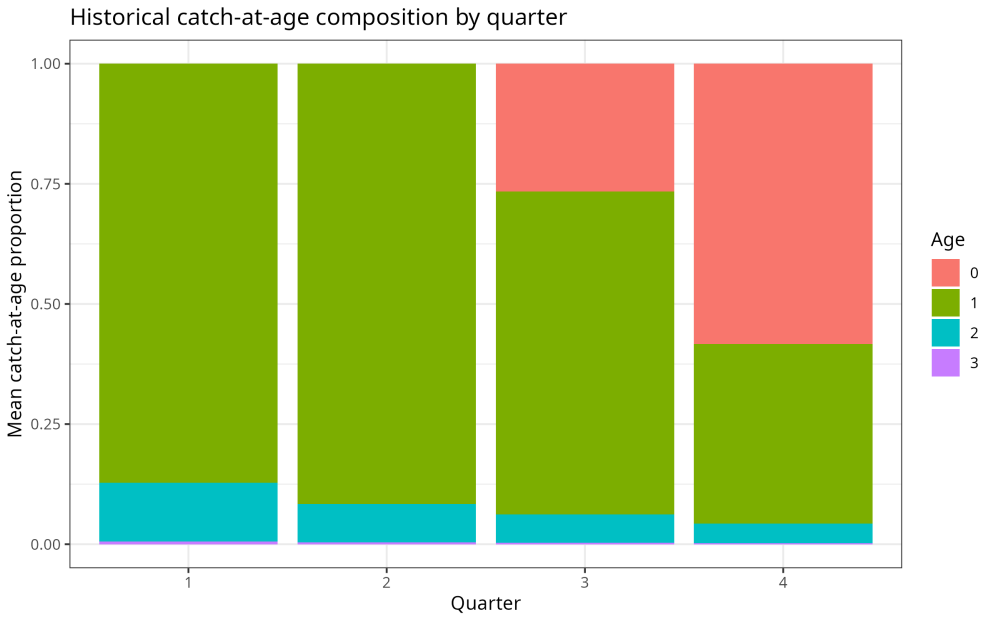


Figure 4: Mean historical catch composition by age and quarter across conditioned OMs. Catch at age is a historical diagnostic; future realized catch at age is generated dynamically during FLBEIA simulations.

4 Reproducibility

The reference fishery is constructed using `10_built_fleet_Reference.R` after generating the biological OM and recruitment-conditioning objects. The script reads the conditioned biological and stock objects, the 1,000-iteration map, and the CSV files containing SEINE fleet variables. It generates:

- `data/Rdata/ane_stock_conditioned_1000iter.rds`;
- `data/Rdata/fleets_conditioned_1000iter.rds`;
- `data/Rdata/fleets_ctrl_reference_1000iter.rds`.

Subsequently, `11_seasonal_catch_allocation_diagnostics.R` reads the conditioned stock, fleet, fleet controls, and iteration map. Reference documentation outputs are stored in `outputs/fishery_conditioning/`, whereas temporal diagnostics, PCA, clustering, and candidate scenarios are stored separately in `outputs/fishery_sensitivity/`.

The main reference settings are summarized in Table 7.

Table 7: Main settings used to construct the reference OM fishery component.

Component	Reference_setting
Historical period	1989-2024
Projection period	2025-2054
Fleet / metier / stock	SEINE / ALL / ANE
Historical OM conditions	100
Projection trajectories	1000 (10 per historical OM)
Effort model	SMFB
Catch model	Baranov
Effort restriction	catch (ANE)
Capital model	fixedCapital
Catch threshold	0.9
Catchability averaging period	2020-2024
Projection weight/fleet averaging period	2022-2024
Reference seasonal-share averaging period	2015-2024, OM-specific
landings.sel	1
discards.sel	0
discard.TAC.OS	FALSE
Future landings.n / discards.n	initialised at 0
Future beta / effort	initialised at 1
Future realised catch-at-age	generated dynamically during simulation

5 Conclusions

- The fishery was incorporated into the OM while maintaining the correspondence between each biological condition and the fishery history derived from the same SS3 bootstrap refit. During the projection, effort, fishing mortality, and catches are generated dynamically in FLBEIA.
- The reference configuration allocates most annual catch to Q2 and Q3, which together account for approximately 75% of the catch. This recent pattern shows little variation among the 100 OMs and is retained during 2025–2054 on an OM-specific basis.
- Seasonal allocation was not constant during 1989–2024. The main changes were observed in Q1 and Q3, with a decrease in the contribution of Q1 and an increase in Q3. Multivariate analyses identified different historical patterns in the distribution of catch among quarters.
- Catch age composition also shows a seasonal structure. Q1 and Q2 catches are dominated by age-1 individuals, whereas the contribution of age 0 increases in Q3 and constitutes the largest proportion of catch in Q4. This historical composition is not imposed during the projection.
- Historical variability in seasonal allocation was used to define alternative exploitation patterns. The patterns corresponding to clusters 1 and 2 were selected as S-C1 and S-C2 to subsequently evaluate MSE sensitivity to changes in the within-year temporal distribution of exploitation.

References

Garcia, D., Sánchez, S., Prellezo, R., Urtizberea, A., and Andrés, M. 2017. FLBEIA: A simulation model to conduct bio-economic evaluation of fisheries management strategies. SoftwareX, 6: 141–147. https://doi.org/10.1016/j.softx.2017.06.001.